Lecture 3: CONVEX PROBLEMS

Phuong L. Vo

International University

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Outline

1. Optimization problems
2. Equivalent problems
3. Convex optimization problems

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3. Convex optimization problems

Optimization problem

Optimization in standard form:

$$\begin{aligned} \min.\ & f_0(x) \\ \text{s.t.}\ & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p. \end{aligned}$$

- $x \in \mathbb{R}^n$: optimization variable,
- $f_0 : \mathbb{R}^n \to \mathbb{R}$: objective function or cost function,
- domain of the optimization problem

$$\mathcal{D} = (\cap_{i=0}^{m} \mathbf{dom} f_i) \cap (\cap_{i=1}^{p} \mathbf{dom} h_i)$$

- feasible set:
  $\{x | x \in \mathcal{D}; f_i(x) \leq 0, i = 1, \ldots, m; h_i(x) = 0, i = 1, \ldots, p\}$

Optimal and locally optimal points

- Optimal value
  $p^* = \min\{f_0(x) | f_i(x) \leq 0, i = 1, \ldots, m, h_i(x) = 0, i = 1, \ldots, p\}$,
- If the problem is unbounded below, then $p^* = -\infty$
  If the problem is infeasible, then $p^* = \infty$
- x^* is an optimal point if x^* is feasible and $f_0(x^*) = p^*$
- The set of all optimal points is called optimal set

$$X_{opt} = \{x | x \in \mathcal{D} \text{ and } f_0(x) = p^*\}$$

- A feasible point x^* is locally optimal if there is an $\epsilon > 0$ such that

$$f_0(x^*) = \min\{f_0(x) | f_i(x) \leq 0, i = 1, \ldots, m, \\ h_i(x) = 0, i = 1, \ldots, p, \|x - x^*\|_2 \leq \epsilon\},$$

  x^* minimizes f_0 over nearby points in the feasible set

Expressing problems in standard form: Problems with box constraints

$$\begin{aligned} \text{min. } & f_0(x) \\ \text{s.t. } & l_i \leq x_i \leq u_i, i = 1, \ldots, n. \end{aligned}$$

Transform to standard form:

$$\begin{aligned} \text{min. } & f_0(x) \\ \text{s.t. } & l_i - x_i \leq 0, i = 1, \ldots, n \\ & x_i - u_i \leq 0, i = 1, \ldots, n. \end{aligned}$$

Expressing problems in standard form: Maximization problems

$$\begin{aligned} \max. \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p. \end{aligned}$$

- Minimizing $-f_0(x)$ equivalently.
- Optimal value:
  $p^* = \max\{f_0(x) | f_i(x) \leq 0, i = 1, \ldots, m, h_i(x) = 0, i = 1, \ldots, p\}$

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Equivalent problems

Two problems are **equivalent** if from a solution of one, a solution of the other is readily found, and vice versa.

For example, the scaled problem is equivalent to the problem in standard form

$$\begin{aligned} \text{min.}\ & \tilde{f}(x) = \alpha_0 f_0(x) \\ \text{s.t}\ & \tilde{f}_i(x) = \alpha_i f_i(x) \leq 0, i = 1, \ldots, m \\ & \tilde{h}_i(x) = \beta_i h_i(x) = 0, i = 1, \ldots, p, \end{aligned}$$

in which $\alpha_i > 0, i = 0, \ldots, m$ and $\beta_i \neq 0, i = 1, \ldots, p$.

Change of variables

Suppose function $\phi : \mathbb{R}^n \to \mathbb{R}^n$ is one-to-one mapping. Define the following functions:

$$\tilde{f}_i(z) = f_i(\phi(z)), i = 0, \ldots, m, \quad \tilde{h}_i(z) = h_i(\phi(z)), i = 0, \ldots, p.$$

By changing the variable $x = \phi(z)$, the following problem is equivalent to the original problem:

$$\begin{aligned} \min. \ & \tilde{f}_0(z) \\ \text{st.} \ & \tilde{f}_i(z) \leq 0, i = 1, \ldots, m \\ & \tilde{h}_i(z) = 0, i = 1, \ldots, p \end{aligned}$$

Transformation of objective and constraint functions

Suppose that $\psi_0 : \mathbb{R}^n \to \mathbb{R}^n$ is monotone increasing, $\psi_1, \ldots, \psi_m : \mathbb{R} \to \mathbb{R}$ satisfy $\psi_i(u) \leq 0$ if and only if $u \leq 0$, and $\xi_1, \ldots, \xi_p : \mathbb{R} \to \mathbb{R}$ satisfy $\xi_i(u) = 0$ if and only if $u = 0$.
Define the following functions:

$$\tilde{f}_i(x) = \psi_i(f_i(x)), i = 0, \ldots, m, \quad \tilde{h}_i(x) = \xi_i(h_i(x)), i = 0, \ldots, p.$$

The following problem is equivalent to the original problem:

$$\begin{aligned} \text{min.} \ & \tilde{f}_0(x) \\ \text{s.t.} \ & \tilde{f}_i(x) \leq 0, i = 1, \ldots, m \\ & \tilde{h}_i(x) = 0, i = 1, \ldots, p. \end{aligned}$$

Examples - Least-norm and least-norm-squared problems

The least-norm and least-norm-square problems are equivalent:

$$\min \ \|Ax - b\|_2 \quad \text{and} \quad \min \ \|Ax - b\|_2^2 = (Ax - b)^T(Ax - b).$$

The optimal points are the same.

Slack variables

By adding slack variables to the inequality constraints, the following problem is equivalent to the original one.

$$
\begin{aligned}
\text{min.}\ & f_0(x) \\
\text{s.t.}\ & s_i \geq 0, i = 1, \ldots, m \\
& f_i(x) + s_i = 0, i = 1, \ldots, m \\
& h_i(x) = 0, i = 1, \ldots, p.
\end{aligned}
$$

Eliminating equality constraints

If we can explicitly parametrize all solutions of the equality constraints $h_i(x) = 0, i = 1, \ldots, p$, i.e., there is function ϕ such that $h_i(x) = 0, i = 1, \ldots, p$ if and only if there is some z such that $x = \phi(z)$ then the following problem is equivalent to the original one.

$$\begin{aligned} &\text{minimization } \tilde{f}(z) = f_0(\phi(z)) \\ &\text{subject to } \tilde{f}_i(z) = f_i(\phi(z)) \leq 0, i = 1, \ldots, m. \end{aligned}$$

Introducing equality constraints

These two problems are equivalent:

$$\begin{aligned} \text{min.} \quad & f_0(A_0x + b_0) \\ \text{s.t.} \quad & f_i(A_ix + b_i) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p. \end{aligned}$$

and

$$\begin{aligned} \text{min.} \quad & f_0(y_0) \\ \text{s.t.} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & y_i = A_ix + b_i, i = 0, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p. \end{aligned}$$

Optimizing over some variables

$$\min_{x_1,x_2} f(x_1,x_2) = \min_{x_1} \tilde{f}(x_1),$$

where $\tilde{f}(x_1) = \min_{x_2} f(x_1,x_2)$.

$$\begin{aligned} \min. \ & f_0(x_1,x_2) \\ \text{s.t.} \ & f_i(x_1) \leq 0, i = 1,\ldots,m_1 \\ & g_i(x_2) \leq 0, i = 1,\ldots,m_2 \end{aligned}$$

is equivalent to

$$\begin{aligned} \min. \ & \tilde{f}_0(x_1) \\ \text{s.t.} \ & f_i(x_1) \leq 0, i = 1,\ldots,m_1 \end{aligned}$$

where $\tilde{f}_0(x_1) = \min_{x_2 | g_i(x_2) \leq 0, i=1,\ldots,m_2} f_0(x_1,x_2)$

Optimizing over some variables

Example 4.4 *Minimizing a quadratic function with constraints on some variables.* Consider a problem with strictly convex quadratic objective, with some of the variables unconstrained:

$$\begin{array}{ll} \text{minimize} & x_1^T P_{11} x_1 + 2x_1^T P_{12} x_2 + x_2^T P_{22} x_2 \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \ldots, m, \end{array}$$

where P_{11} and P_{22} are symmetric. Here we can analytically minimize over x_2:

$$\inf_{x_2} \left(x_1^T P_{11} x_1 + 2x_1^T P_{12} x_2 + x_2^T P_{22} x_2 \right) = x_1^T \left(P_{11} - P_{12} P_{22}^{-1} P_{12}^T \right) x_1$$

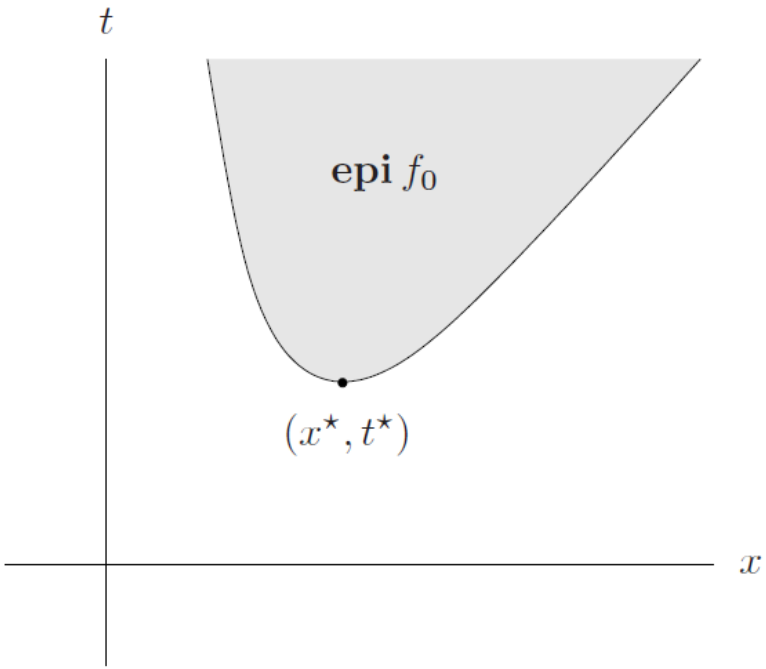
(see §A.5.5). Therefore the original problem is equivalent to

$$\begin{array}{ll} \text{minimize} & x_1^T \left(P_{11} - P_{12} P_{22}^{-1} P_{12}^T \right) x_1 \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \ldots, m. \end{array}$$

Epigraph form

The problem in epigraph form is equivalent to the original problem.

$$\begin{aligned} \text{min. } & t \\ \text{s.t. } & f_0(x) - t \leq 0 \\ & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p. \end{aligned}$$



(x^*, t^*) is optimal for epigraph problem if and only if x^* is optimal for original and $t^* = f_0(x^*)$.

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Convex optimization problems in standard form

$$\begin{aligned} \text{min.} \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & a_i^T x = b_i, \quad i = 1, \ldots, p. \end{aligned}$$

- the objective function must be convex,
- the inequality constraint functions must be convex,
- the equality constraint functions $h_i(x) = a_i^T x - b_i$ must be affine.

The feasible set of a convex optimization problem is convex (Why ?)

Concave maximization problems

$$\begin{aligned} \max.\ & f_0(x) \\ \text{s.t.}\ & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & a_i^T x = b_i, \quad i = 1, \ldots, p. \end{aligned}$$

The problem is convex optimization problem is the objective f_0 is concave and the inequality constraint functions $f_1, \ldots, f_m$ are convex.

Local and global optima

Any locally optimal point of a convex problem is also (globally) optimal.
(See page 138 for the proof.)

Exercise

Problem 4.1 page 189

An optimality criterion for differentiable f_0

Define the feasible set
$X = \{x | f_i(x) \leq 0, i = 1, \ldots, m, h_i(x) = 0, i = 1, \ldots, p\}$. x^* is optimal if and only if $x^* \in X$ and

$$\nabla f_0(x^*)^T (y - x^*) \geq 0 \quad \forall y \in X.$$

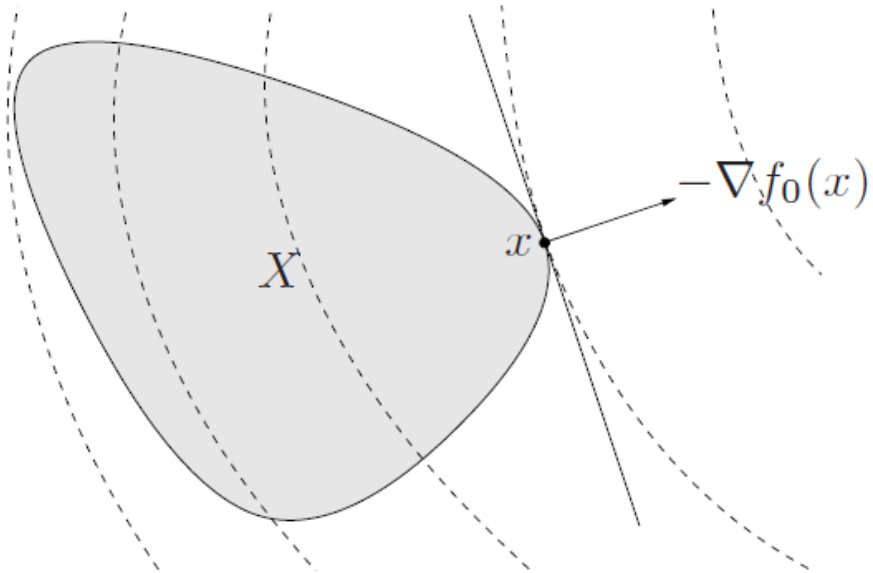


Figure: The feasible set X is shown shaded. Some level curves of f_0 are shown as dashed lines. The point x is optimal: $-\nabla f_0(x)$ defines a supporting hyperplane (shown as a solid line) to X at x.

An optimality criterion for differentiable f_0

For unconstrained problems, necessary and sufficient condition for x optimal is:

$$\nabla f_0(x) = 0$$

Examples 4.5, page 140.

Linear program

$$\begin{aligned} \text{min. } & c^T x + d \\ \text{s.t. } & Gx \leq h \\ & Ax = b, \end{aligned}$$

where $G \in \mathbb{R}^{m \times n}$ and $A \in \mathbb{R}^{p \times n}$.

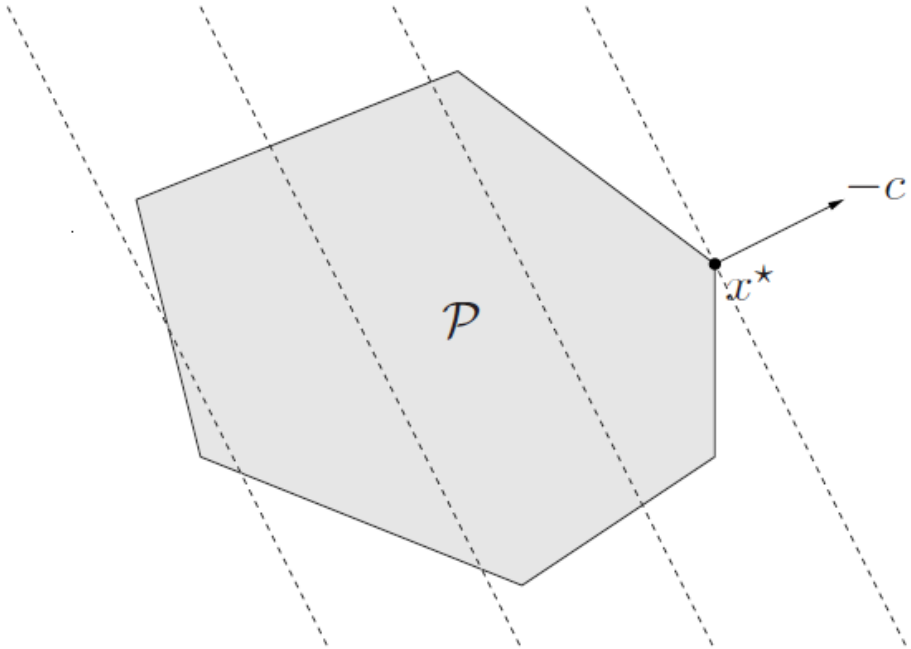


Figure: Geometric interpretation of an LP. The feasible set P, which is a polyhedron, is shaded. The objective $c^T x$ is linear, so its level curves are hyperplanes orthogonal to c (shown as dashed lines). The point x^* is optimal; it is the point in P as far as possible in the direction $-c$.

Quadratic optimization problems

$$\begin{aligned} \text{min.} \quad & (1/2)x^T Px + q^T x + r \\ \text{s.t.} \quad & Gx \leq h \\ & Ax = b, \end{aligned}$$

where $P \in \mathbb{R}^{n \times n}$, $P \geq 0$, $G \in \mathbb{R}^{m \times n}$ and $A \in \mathbb{R}^{p \times n}$.

Quadratic optimization problems: Least-squares and regression

Minimizing the function in an unconstrained QP.

$$\|Ax - b\|_2^2 = x^T A^T Ax - 2b^T Ax + b^T b.$$

- analytical solution $x^* = A^\dagger b$, where $A^\dagger = (A^T A)^{-1} A^T$ is the pseudo-inverse matrix of A
- can add linear constraints, e.g., $l \leq x \leq u$

Quadratic optimization problems: Distance between polyhedra

Define **dist**$(\mathcal{P}_1, \mathcal{P}_2) = \min \|x_1 - x_2\|_2$,
$x_1 \in \mathcal{P}_1 = \{x | A_1 x \leq b_1\}, x_2 \in \mathcal{P}_2 = \{x | A_2 x \leq b_2\}$. The problem is as follows:

$$\begin{aligned} \min \ & \|x_1 - x_2\|_2^2 \\ \text{s.t.} \ & A_1 x_1 \leq b_1, \\ & A_2 x_2 \leq b_2. \end{aligned}$$

Quadratic optimization problems: Linear program with random cost (portfolio optimization)

$$\begin{aligned} \text{min.} \quad & \bar{c}^T x + \gamma x^T V x \\ \text{s.t.} \quad & Gx \leq h \\ & Ax = b. \end{aligned}$$

- Cost function $c \in \mathbb{R}^n$ is random vector with mean $\bar{c} = \mathbb{E}[c]$ and covariance $V = \mathbb{E}(c - \bar{c})(c - \bar{c})^T$.
- Given x, $c^T x$ is a random variable with mean $\bar{c}^T x$ and variance $x^T V x$.
- $\gamma > 0$ is risk aversion parameter; controls the trade-off between expected cost and variance (risk).

Geometric programming

- Monomial function: $f(x) = cx_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$, where $c > 0, a_i \in \mathbb{R}$.
- Posynomial function: $f(x) = \sum_{k=1}^{K} cx_1^{a_{1k}}x_2^{a_{2k}}\ldots x_n^{a_{nk}}$ where $c_k > 0$.
- Optimization problem in the form

$$\begin{aligned} \text{min. } & f_0(x) \\ \text{s.t. } & f_i(x) \leq 1, \quad i = 1, \ldots, m \\ & h_i(x) = 1, \quad i = 1, \ldots, p, \end{aligned}$$

where $f_0, \ldots, f_m$ posynomial, h_i monomial.

Geometric programming

Change variables to $y_i = \log x_i$, and take logarithm of cost, constraints

- monomial $f(x) = cx_1^{a_1} x_2^{a_2} \dots x_n^{a_n}$ transforms to

$$\log f(e^{y_1}, \dots, e^{y_n}) = a^T y + b \quad (b = \log c)$$

- posynomial $f(x) = \sum_{k=1}^{K} cx_1^{a_{1k}} x_2^{a_{2k}} \dots x_n^{a_{nk}}$ transforms to

$$\log f(e^{y_1}, \dots, e^{y_n}) = \log \left(\sum_{i=1}^{K} exp(a_{ik}^T + b_{ik}) \right) \quad (b_k = \log c_k)$$

- geometric program transforms to convex problem

$$\begin{aligned} \max. \ & \log \left(\sum_{i=1}^{K} exp(a_{0k}^T + b_{0k}) \right) \\ \text{s.t.} \ & \log \left(\sum_{i=1}^{K} exp(a_{ik}^T + b_{ik}) \right) \leq 1, \quad i = 1, \dots, m \\ & Gy + d = 0, \quad i = 1, \dots, p, \end{aligned}$$

Quadratically constrained quadratic program (QCQP)

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

- $P_i \in \mathbf{S}_+^n$; objective and constraints are convex quadratic
- if $P_1, \ldots, P_m \in \mathbf{S}_{++}^n$, feasible region is intersection of m ellipsoids and an affine set

Second-order cone programming

$$\begin{array}{ll} \text{minimize} & f^T x \\ \text{subject to} & \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \ldots, m \\ & Fx = g \end{array}$$

$(A_i \in \mathbf{R}^{n_i \times n},\ F \in \mathbf{R}^{p \times n})$

- inequalities are called second-order cone (SOC) constraints:

$$(A_i x + b_i, c_i^T x + d_i) \in \text{second-order cone in } \mathbf{R}^{n_i+1}$$

- for $n_i = 0$, reduces to an LP; if $c_i = 0$, reduces to a QCQP
- more general than QCQP and LP

Semidefinite program (SDP)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & x_1 F_1 + x_2 F_2 + \cdots + x_n F_n + G \preceq 0 \\ & Ax = b \end{array}$$

with F_i, $G \in \mathbf{S}^k$

- inequality constraint is called linear matrix inequality (LMI)
- includes problems with multiple LMI constraints: for example,

$$x_1 \hat{F}_1 + \cdots + x_n \hat{F}_n + \hat{G} \preceq 0, \qquad x_1 \tilde{F}_1 + \cdots + x_n \tilde{F}_n + \tilde{G} \preceq 0$$

is equivalent to single LMI

$$x_1 \begin{bmatrix} \hat{F}_1 & 0 \\ 0 & \tilde{F}_1 \end{bmatrix} + x_2 \begin{bmatrix} \hat{F}_2 & 0 \\ 0 & \tilde{F}_2 \end{bmatrix} + \cdots + x_n \begin{bmatrix} \hat{F}_n & 0 \\ 0 & \tilde{F}_n \end{bmatrix} + \begin{bmatrix} \hat{G} & 0 \\ 0 & \tilde{G} \end{bmatrix} \preceq 0$$

LP and SOCP as SDP

LP and equivalent SDP

$$\begin{array}{lllll} \text{LP:} & \text{minimize} & c^T x & \text{SDP:} & \text{minimize} \quad c^T x \\ & \text{subject to} & Ax \preceq b & & \text{subject to} \quad \mathbf{diag}(Ax - b) \preceq 0 \end{array}$$

(note different interpretation of generalized inequality $\preceq$)

SOCP and equivalent SDP

$$\begin{array}{lll} \text{SOCP:} & \text{minimize} & f^T x \\ & \text{subject to} & \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \ldots, m \end{array}$$

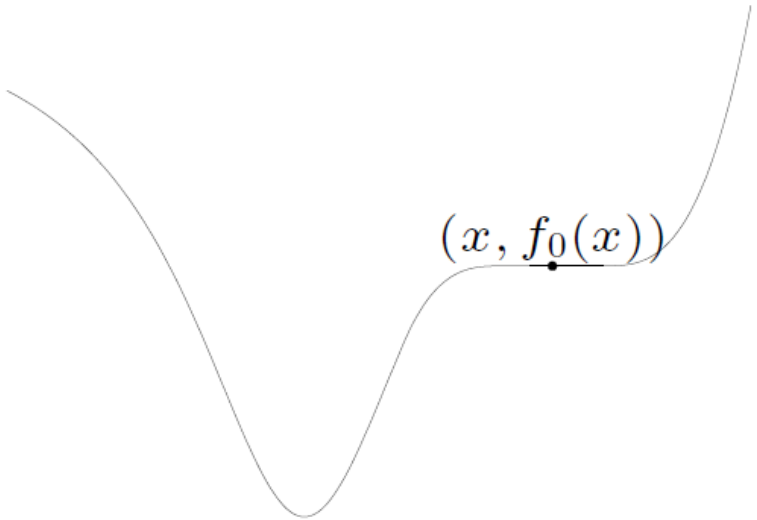
$$\begin{array}{lll} \text{SDP:} & \text{minimize} & f^T x \\ & \text{subject to} & \begin{bmatrix} (c_i^T x + d_i) I & A_i x + b_i \\ (A_i x + b_i)^T & c_i^T x + d_i \end{bmatrix} \succeq 0, \quad i = 1, \ldots, m \end{array}$$

Quasiconvex optimization

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

with $f_0 : \mathbf{R}^n \to \mathbf{R}$ quasiconvex, $f_1, \ldots, f_m$ convex

can have locally optimal points that are not (globally) optimal



Thank you for your attention.